

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

Paper 2

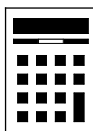
Monday 20 January 2025 07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
5	
6	
7	
8	
TOTAL	



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9202/2

Answer **all** questions in the spaces provided.

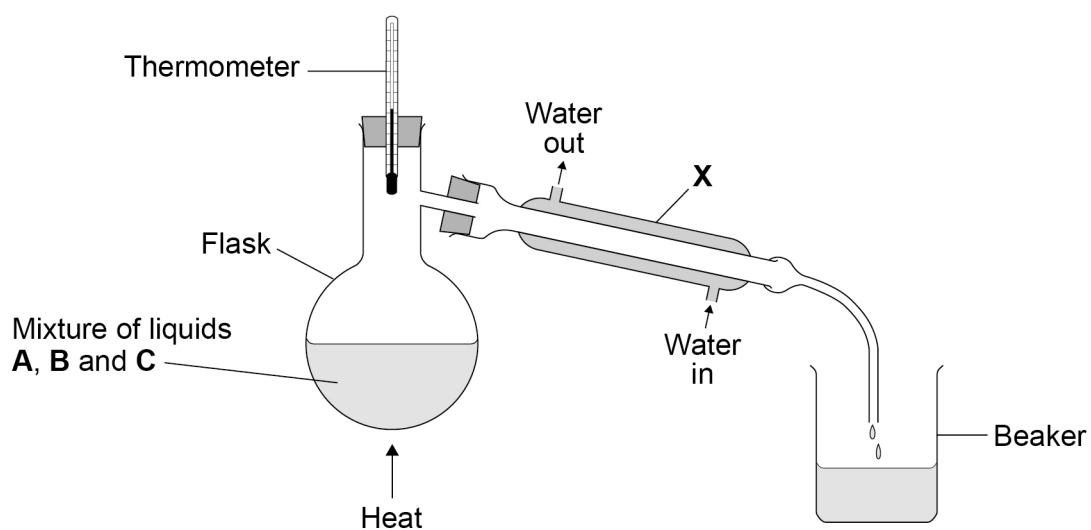
0 1

This question is about separating substances in mixtures.

A student separated a mixture of liquids.

Figure 1 shows the apparatus.

Figure 1



0 1 . 1

What is the process shown in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Crystallisation

☐

Distillation

☐

Electrolysis

☐

Filtration

☐

0 1 . 2 What is the name of the piece of equipment labelled **X** in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Condenser

☐

Conical flask

☐

Filter funnel

☐

Measuring cylinder

☐

0 1 . 3 **Table 1** shows the boiling points of the liquids in the mixture used in **Figure 1**.

Table 1

Liquid	Boiling point in °C
A	78
B	56
C	100

The mixture of liquids in **Figure 1** is slowly heated.

Which liquid would collect in the beaker first?

Give **one** reason why.

[2 marks]

Liquid _____

Reason _____

Question 1 continues on the next page

Turn over ►



0 1 . 4

The thermometer always gave a reading 3 °C higher than the true value.

What type of error was this?

[1 mark]

Tick (✓) **one** box.

Measurement error

☐

Random error

☐

Systematic error

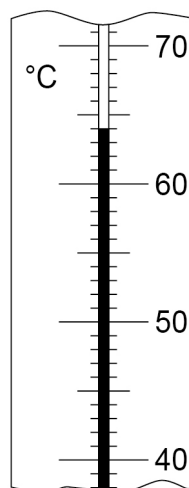
☐

The student replaced the thermometer.

The new thermometer did **not** have an error.

Figure 2 shows a section of the new thermometer after heating the liquids.

Figure 2



0 1 . 5

What is the temperature shown on the thermometer in **Figure 2**?

[1 mark]

_____ °C



0 1 . 6

What is the **resolution** of the thermometer in **Figure 2**?**[1 mark]**Tick (✓) **one** box.

1 °C

☐

5 °C

☐

10 °C

☐

60 °C

☐

0 1 . 7

A different mixture contains an insoluble solid and a salt solution.

Draw the apparatus used to separate an insoluble solid from a solution.

[2 marks]**Turn over ►**

0 2

This question is about Group 7 elements and the compounds of Group 7 elements.

0 2 . 1

What is the name of the Group 7 elements?

[1 mark]Tick (✓) **one** box.

Alkali metals

☐

Halogens

☐

Noble gases

☐

Transition metals

☐**0 2 . 2**

Fluorine and chlorine are Group 7 elements.

Why are fluorine and chlorine in the same group?

[1 mark]Tick (✓) **one** box.

Both have the same atomic number.

☐

Both have the same number of neutrons.

☐

Both have the same number of outer electrons.

☐

Both have the same number of protons.

☐

Table 2 shows the melting points of some Group 7 elements.

Table 2

Element	Melting point in °C
Fluorine	-220
Chlorine	X
Bromine	-7
Iodine	114

0 2 . 3 Predict the value of **X** in **Table 2**.

[1 mark]

_____ °C

0 2 . 4 What is the state of iodine at 25 °C?

Use **Table 2**.

[1 mark]

0 2 . 5 Bromine reacts with potassium iodide solution.

What type of reaction occurs when bromine reacts with potassium iodide solution?

[1 mark]

Tick (✓) **one** box.

Combustion

☐

Decomposition

☐

Displacement

☐

Neutralisation

☐

Question 2 continues on the next page

Turn over ►



0 2 . 6

A potassium atom reacts with a chlorine atom to form a potassium ion and a chloride ion.

Figure 3 shows the electronic structure of a potassium atom and a chlorine atom.

Only the outer electrons are shown.

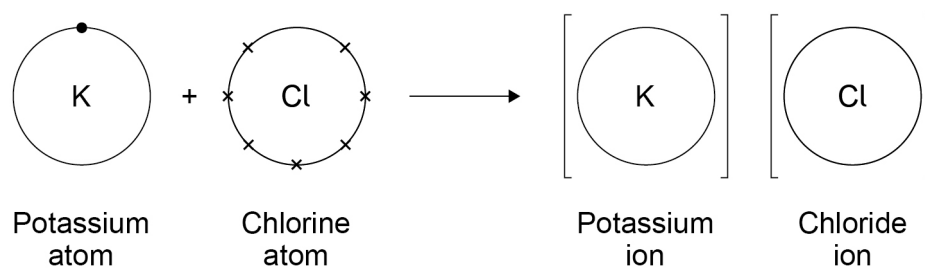
Complete **Figure 3**.

You should include:

- all outer shell electrons on the ions
- the charge on each ion.

[4 marks]

Figure 3



Silver chloride is a compound of silver and chlorine.

Silver chloride nanoparticles can be used to reduce the number of bacteria on surfaces.

Table 3 shows information about three different silver chloride nanoparticles, **A**, **B** and **C**.

Table 3

Silver chloride nanoparticle	Surface area in nm ²	Volume in nm ³	Surface area:volume ratio	Percentage reduction in bacteria in %
A	600	1 000	0.6:1	80
B	15 000	125 000	X	45
C	38 400	512 000	0.075:1	25

0 2 . 7

What is the trend between the surface area of the silver chloride nanoparticle and the percentage reduction in bacteria?

[1 mark]

Tick (✓) **one** box.

As the surface area increases, the percentage reduction in bacteria decreases.

☐

Surface area has no effect on the percentage reduction in bacteria.

☐

As the surface area increases, the percentage reduction in bacteria increases.

☐

0 2 . 8

Determine the surface area to volume ratio (**X**) for nanoparticle **B**.

Use **Table 3**.

[2 marks]

Surface area:volume ratio (**X**) = _____ :1

12

Turn over ►



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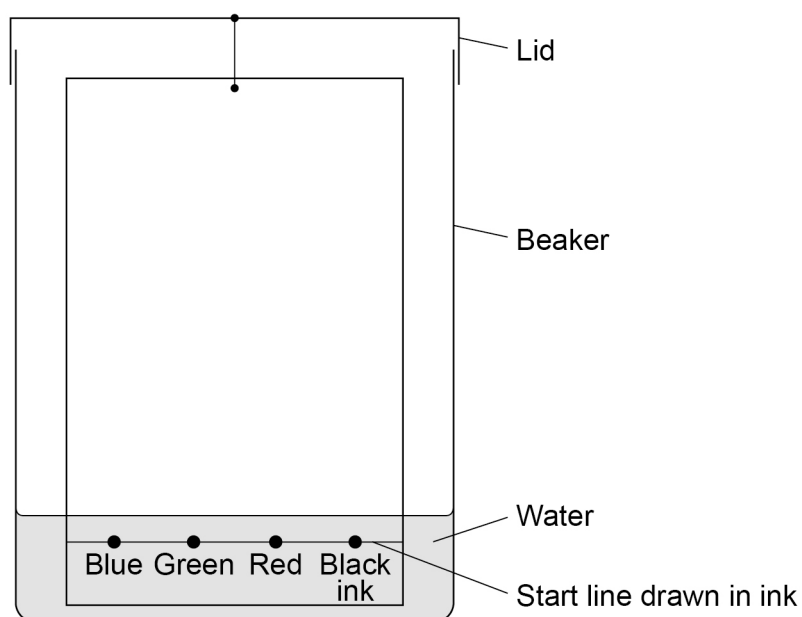
0 3

A student analysed the colours in black ink using chromatography.

0 3 . 1

Figure 4 shows the apparatus.

Figure 4



Give **two** errors made by the student.

Give **one** improvement for each error.

[4 marks]

Error 1 _____

Improvement 1 _____

Error 2 _____

Improvement 2 _____

Question 3 continues on the next page

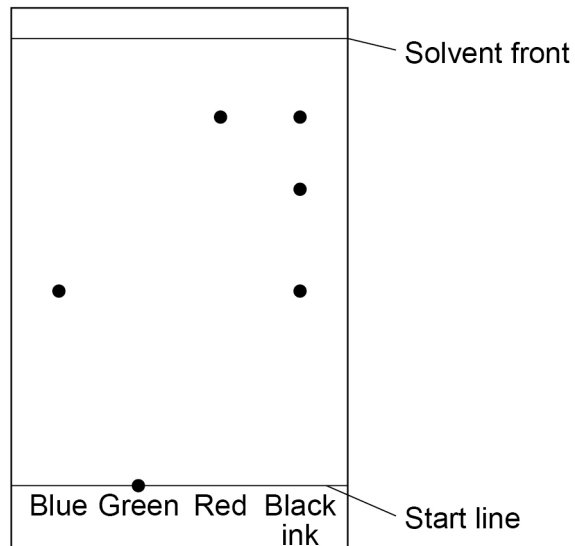
Turn over ►



A different student used the correct apparatus.

Figure 5 shows the results.

Figure 5



0 3 . 2

Give **three** conclusions that can be made about the black ink.

Use **Figure 5**.

[3 marks]

1 _____

2 _____

3 _____



0 3 . 3 Determine the R_f value of the **red colour**.

Use **Figure 5**.

[4 marks]

R_f value of red colour = _____

0 3 . 4 The student then investigated four different inks using chromatography.

Table 4 shows the results.

Table 4

Ink	R_f value
Fast violet	0.90
Light blue	0.64
Sky blue	0.65
Yellow	0.79

Which ink is **most soluble** in the mobile phase?

Give **one** reason for your answer.

[2 marks]

Ink _____

Reason _____



0 5

A student investigated the reaction between sodium thiosulfate solution and hydrochloric acid.

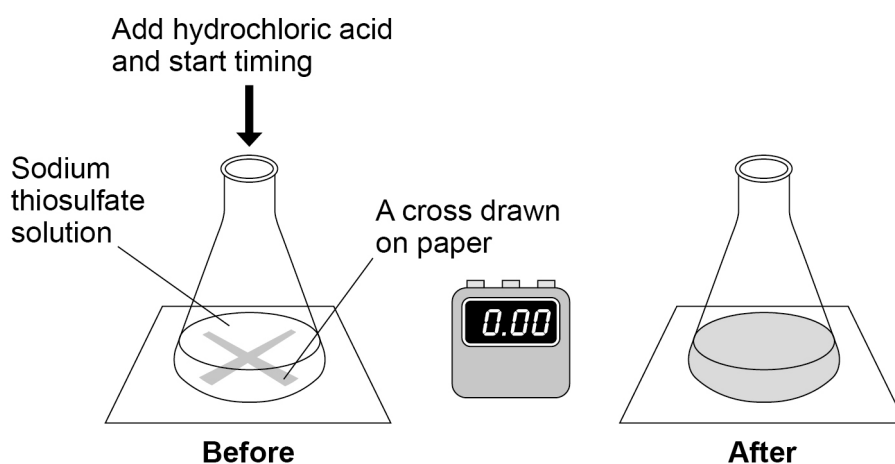
The student changed the concentration of the sodium thiosulfate solution to see the effect on the rate of reaction.

This is the method used:

- 1 Measure 50 cm³ of sodium thiosulfate solution into a conical flask.
- 2 Draw a cross on paper.
- 3 Place the conical flask on the cross.
- 4 Measure 5 cm³ of hydrochloric acid.
- 5 Pour the hydrochloric acid into the conical flask and immediately start the timer.
- 6 Stop the timer when the cross is no longer visible.
- 7 Repeat steps 1–6 with different concentrations of sodium thiosulfate solution.

Figure 7 shows the apparatus before and after the reaction.

Figure 7



The equation for the reaction is:



0 5

1

Explain why the cross is no longer visible.

[2 marks]



Table 5 shows the results.

Table 5

Concentration of sodium thiosulfate solution in g/dm ³	Time for cross to no longer be visible in seconds			Mean time for cross to no longer be visible in seconds
	Test 1	Test 2	Test 3	
8	81	80	82	81
16	42	47	46	45
24	31	X	38	34
32	25	24	28	26
40	22	22	25	23

0 5 . 2 Calculate value **X** in **Table 5**.

[3 marks]

X = _____ s

0 5 . 3 The student states that the results for the 8 g/dm³ concentration of sodium thiosulfate solution are **precise**.

The student is correct.

Give **one** reason why these results are precise.

Use **Table 5**.

[1 mark]

Turn over ►



Table 5 is repeated below.

Table 5

Concentration of sodium thiosulfate solution in g/dm ³	Time for cross to no longer be visible in seconds			Mean time for cross to no longer be visible in seconds
	Test 1	Test 2	Test 3	
8	81	80	82	81
16	42	47	46	45
24	31	X	38	34
32	25	24	28	26
40	22	22	25	23

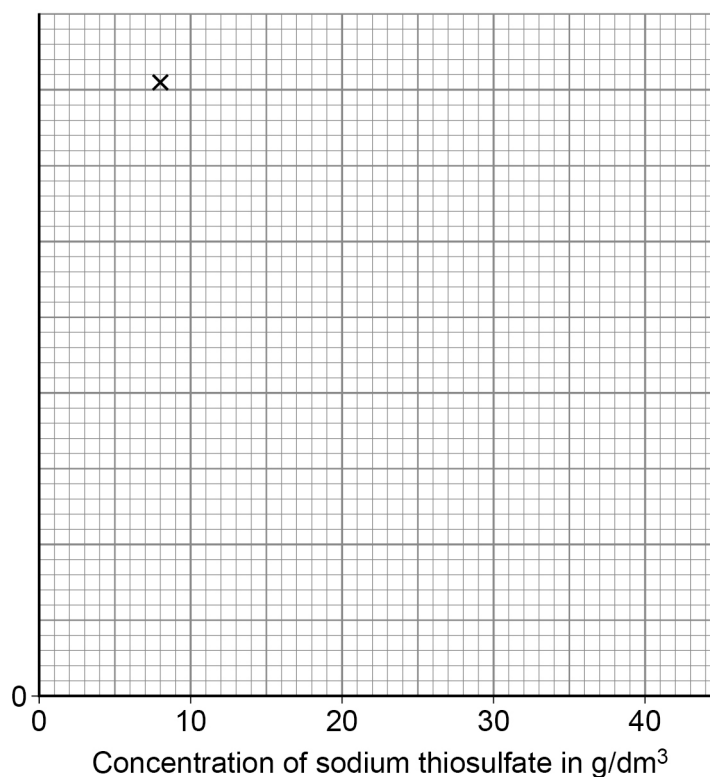


0 5 . 4

Figure 8 shows an incomplete graph of the results in Table 5.

Figure 8

Mean time
for cross to
no longer
be visible
in seconds



One point has been plotted for you.

Complete **Figure 8**.

You should:

- use a suitable scale for the y -axis
- plot the data from **Table 5**
- draw a line of best fit.

[4 marks]

0 5 . 5

Describe the trend shown in **Figure 8**.

[1 mark]

Question 5 continues on the next page

Turn over ►



0 5 . 6

Explain how increasing concentration affects the rate of reaction.

[3 marks]

14

[6 marks]

0 6 . 2

The student tested an aqueous solution of compound **X**.

Table 6 shows the tests and the observations made.

Table 6

Test	Observation
Flame test	Lilac flame
Add hydrochloric acid followed by barium chloride	White precipitate

Name compound **X**.

[2 marks]

8



0 7

Sodium hydroxide solution is neutralised by sulfuric acid to produce a salt and water.

0 7 . 1

What is the pH of a neutral solution?

[1 mark]Tick (✓) **one** box.

2

☐

7

☐

9

☐

14

☐**0 7 . 2**

Which salt is produced when sulfuric acid reacts with sodium hydroxide solution?

[1 mark]Tick (✓) **one** box.

Sodium chloride

☐

Sodium nitrate

☐

Sodium sulfate

☐**0 7 . 3**

Complete the ionic equation for neutralisation reactions.

You should include state symbols.

[2 marks]**Question 7 continues on the next page****Turn over ►**

The student used:

- a solution of sodium hydroxide of unknown concentration
- sulfuric acid of concentration 0.200 mol/dm^3 .

Describe a method the student could use to determine the concentration of the solution of sodium hydroxide.

You do **not** need to explain how to calculate the concentration.

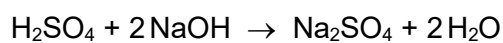
[6 marks]

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0	7	.	5
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28.10 cm³ of 0.200 mol/dm³ sulfuric acid neutralised 25.0 cm³ of the sodium hydroxide solution.

The equation for the reaction is:



Calculate the concentration of the sodium hydroxide solution.

Give your answer to 3 significant figures.

[5 marks]

Concentration (3 significant figures) = _____ mol/dm³

15

Turn over for the next question

Turn over ►



0 8

This question is about the biofuel ethanol.

0 8 . 1

Petrol is a hydrocarbon fuel.

Both petrol and ethanol can be used to fuel a car.

Table 7 gives information about both fuels.**Table 7**

Fuel	Raw Material	Grams of carbon dioxide released per mile of driving
Ethanol	Plant material	315
Petrol	Crude oil	404

Give **two** environmental advantages of using ethanol rather than petrol as a fuel.**[2 marks]**

- 1 _____
- _____
- 2 _____
- _____

0 8 . 2

Incomplete combustion occurs if not enough oxygen is present.

Complete the equation for the incomplete combustion of ethanol.

You should balance the equation.

[2 marks]

0	8	.	3
---	---	---	---

A solution of ethanol had a concentration of 21.0 g/dm^3 .

Calculate the number of ethanol molecules in 0.250 dm^3 of this solution.

Avogadro's constant = 6.02×10^{23}

Relative formula mass (M_r): ethanol = 46

[5 marks]

Number of ethanol molecules in solution = _____

9

END OF QUESTIONS



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